



P28

Million-Scale Communication Protocols: Engineering Constraints of Replayable Event Streams and Sub-10ms Coordination

百万级通信协议：可重放事件流与亚 10ms

Million-Scale Communication Protocols: Engineering Constraints of Replayable Event Streams and Sub-10ms Coordination

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One sentence first: P1 at the protocol layer proves "the layered tree presses fan-in from linear to constant," but that is a zero-network closed-form count; in production, multi-Agent communication must solve another set of engineering problems—session isolation, checkpoint resume, event replayability, sub-10-millisecond decision latency. RocketMQ-A2A (FSE 2026 Industry) gives production-grade replayable event streams, Adaptive Bitmask gives the sub-10-millisecond direction, and the AI Agent Protocols survey gives a scalability metric (CNS). This paper makes this engineering-constraint layer clear and keeps a downgraded caliber for circulated precise numbers.

Evidence caliber (stated only once in the full text): this paper is an engineering-survey/constraint-analysis paper containing no new UDOS measured numbers. External literature is verified per EVIDENCE_LEDGER_v76.md: RocketMQ-A2A (ACM FSE 2026 Industry, Alibaba Cloud) session-level replayable event streams, LiteTopic session isolation, checkpoint resume are verified facts, "tens-of-millions-level sessions" is product/paper-reported; the AI Agent Protocols survey (arXiv:2504.16736) CNS formula $\text{CNS} = (N_{\text{success}} / N_{\text{attempts}}) / T_{\text{avg}}$ is verified verbatim. Adaptive Bitmask (npm package) sub-10-millisecond direction is true, but "85× compression / 8.2ms p99 / N=5000" is **author claim, not independently rechecked**, and this paper cites it downgraded. Sovereign Negotiator and Ripple Effect Protocol are mentioned only by direction, their specific numbers marked [CITATION NEEDED].

Abstract

P1 used a zero-network discrete-event simulator to prove the layered topology's fan-in/rounds growth order is controllable; but real multi-Agent systems deploy on message middleware, and the communication layer has a set of engineering constraints P1 did not cover: how sessions are isolated, how interruptions recover, how event streams are audited and replayed, how decision latency is pressed to sub-10 milliseconds. This paper reviews the three main threads of this layer in 2026. First, **replayable event streams:** RocketMQ-A2A (FSE 2026 Industry) uses LiteTopic for session-level isolation, checkpoint resume, replayable event streams, solving recovery and audit for large-scale multi-Agent in production. Second, **sub-10-millisecond coordination:** Adaptive Bitmask claims to use semantic bitmasks to press coordination to sub-10 milliseconds, with high-load compression, but its precise numbers (85×/8.2ms p99) are author claims, not independently rechecked. Third, **scalability metrics:** the AI Agent Protocols survey proposes CNS and a node/link scalability framework, making clear a scalable protocol must support thousands to millions of Agents. This paper argues: the communication layer's real bottleneck is not "whether bandwidth suffices" but "whether it can deterministically replay after faults" and "whether coordination latency can beat the decision cycle"; and it connects this layer's constraints back to P1's protocol-layer conclusion. Two falsifiable conditions are given.

Keywords: multi-Agent communication; replayable event streams; session isolation; sub-10-millisecond coordination; protocol scalability; CNS; message middleware

Structured Abstract (Background problem → Argument → Evidence → Contributions)

- **Background problem:** P1's fan-in/rounds are zero-network closed-form counts, not touching production message middleware. Truly deploying a multi-Agent system to production, the communication layer's session isolation, recovery, audit, latency are the hard constraints deciding "whether volume can be taken on."
 - **Core argument:** the bottleneck of million-scale communication is not bandwidth but **deterministic replay after faults** and **coordination latency beating the decision cycle**; replayable event streams (RocketMQ-A2A) solve the former, bitmask/semantic encoding (Adaptive Bitmask) directionally solves the latter.
 - **Evidence threads:** (1) RocketMQ-A2A replayable event streams (FSE 2026 Industry); (2) Adaptive Bitmask sub-10ms direction (precise numbers not rechecked); (3) AI Agent Protocols survey CNS metric (arXiv:2504.16736); (4) P1 protocol-layer fan-in conclusion (cpu-proto).
 - **Contributions:** connecting P1's protocol-layer conclusion to the production communication layer; distinguishing "bandwidth bottleneck" from "replay/latency bottleneck"; downgrading circulated precise numbers; two falsifiable conditions.
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1 Introduction: After P1, One More Layer Is Missing

P1 answered "how Agents organize messages without exploding": at a million nodes the star center takes about 20.97 million messages, the layered tree presses every node's fan-in to constant 9, rounds to 14 (cpu-proto). But this conclusion was obtained in a deterministic discrete-event simulator with **zero network and zero large-model calls**—it assumes messages arrive instantly, not lost, not duplicated.

Production is not so. Messages are lost, reordered, backlogged; after an Agent crashes, which step it processed to, whether to replay, from where to replay—P1 touched none of these. The 2026 multi-Agent communication-protocol work is precisely filling this layer. This paper reviews this layer's three main threads and keeps a downgraded caliber for precise numbers within.

2 Thread One: Replayable Event Streams (RocketMQ-A2A)

2.1 What It Solves

Verified per EVIDENCE_LEDGER_v76.md (2026-09-21), RocketMQ-A2A (RocketMQ-A2A: Reliable Session-Level Replayable Event Streams for Large-Scale Multi-Agent Collaboration, ACM FSE 2026 Industry Papers Track, Alibaba Cloud) is a production-grade solution. Its core is **session-level replayable event streams**:

- **LiteTopic session isolation:** providing independent message isolation per Agent session, avoiding large-scale sessions cross-talking (product/paper reports claim support for tens-of-millions-level session isolation).

- **Checkpoint resume:** interrupted tasks can fully recover from the last checkpoint rather than rerun from scratch.
- **Replayable event streams:** after downstream faults, event streams can be fully replayed and complete audit trails.

2.2 Why "Replayable" Is Key

This is especially important for UDOS. UDOS P1's fault end-to-end tasks rely on hash-chained Trace for audit; RocketMQ-A2A sinks "replayable event streams" from the application layer down to the message-middleware layer—meaning "from which step to recover after faults, how to replay audit" is no longer a patch each application implements itself but a native capability of the communication layer. This is engineering-isomorphic to UDOS P12's evidence grading and P31's drift-baseline rollback: **replayable = auditable = rollbackable**.

Conversely, if the communication layer does not provide replay, the application layer must itself maintain a set of "event log + checkpoint records + state-machine replay," which in large-scale, long-horizon tasks almost necessarily errs—one missed log entry, state machine inconsistent with replay results, both turn "recoverable" into "seemingly recovered." RocketMQ-A2A's value is precisely taking this error-prone piece from the application and handing it to a middleware team specialized in message reliability. This reflects engineering maturity: the closer to the bottom reliability lies, the less it should be reimplemented by business applications.

Figure 1 Three engineering constraints of the communication layer

Figure 1 Session isolation (no cross-talk), checkpoint resume (no restart), event replayability (auditable) are the trio of production-grade multi-Agent communication; RocketMQ-A2A natively provides them at the message-middleware layer. Conceptual illustration.

3 Thread Two: Sub-10-Millisecond Coordination (Adaptive Bitmask)

3.1 Direction True, Numbers Downgraded

Adaptive Bitmask (npm package) claims to encode coordination information with a **semantic bitmask**, pressing the quadratic coordination complexity at $\$N^2\$$ to the $\$O(N)\$$ magnitude and achieving sub-10-millisecond decision latency. Verified per EVIDENCE_LEDGER: the sub-10-millisecond direction is true, but "85× load compression (24B vs 2KB) / 8.2ms p99 / N=5000" is **not independently rechecked**, and this volume downgrades it to "author claim."

3.2 What It Is Solving

Whatever the precise numbers, the engineering problem it points to is real: after the Agent count $\$N\$$ reaches the thousands, point-to-point negotiation traffic is $\$O(N^2)\$$, coordination latency growing quadratically with $\$N\$$. The bitmask/semantic-encoding idea compresses "each Agent separately sending one message" into "one shared bit vector"—using bit operations in place of point-to-point negotiation. This shares a source in thought with P1's layered tree: both **use a more compact representation to replace $\$O(N^2)\$$ negotiation**.

But it must be noted: Adaptive Bitmask's "sub-10 milliseconds" is **coordination decision latency**, not end-to-end task completion time; it compresses the representation of negotiation messages and does not solve the ambiguity of semantic acceptance (P21's extrapolation boundary).

4 Thread Three: Scalability Metrics (AI Agent Protocols Survey)

4.1 CNS and Node/Link Scalability

The AI Agent Protocols survey (arXiv:2504.16736) proposes a protocol-scalability metric framework: node scalability, link scalability, and gives the **Capability Negotiation Score**

$$\text{CNS} = \frac{N_{\text{success}}}{N_{\text{attempts}}} T_{\text{avg}} \tag{1}$$

i.e. the negotiation success rate per unit average time. The survey makes clear: a scalable Agent protocol must support thousands to millions of Agents. This metric turns "whether the protocol is good" from qualitative intuition into a table-lookup quantity—the same methodology as UDOS P1 turning "fan-in" from intuition into closed-form counts.

4.2 Division with P1

P1 measures the fan-in/rounds growth order under a **topology structure**; CNS measures the per-unit-time negotiation success rate of the **protocol implementation**. The two are complementary: P1 tells you the layered tree theoretically presses fan-in to a constant, CNS tells you how many effective negotiations this protocol can complete per unit time in measurement. One is a structural upper bound, one engineering measurement.

4.3 Scalability Is Not "How Many Can Connect," but "Whether It Can Restart After Breaking"

The survey splits scalability into node scalability and link scalability. Beyond these two dimensions, this paper argues for adding a third—**recovery scalability**: after a protocol runs large-scale and long, whether meeting one fault it can deterministically recover to a consistent state without requiring full rerun. RocketMQ-A2A's replayable event streams precisely fill this dimension: node scalability cares "whether more can connect," link scalability "whether connections are good," recovery scalability "whether it can retreat after error."

This dimension is not abstract for UDOS. UDOS P10's TransferBundle six-field handoff contract, P12's evidence grading, P31's drift-baseline rollback essentially all implement at the application layer "verifiable after handoff, rollbackable on problems." RocketMQ-A2A's value lies in sinking this from the application layer to message middleware—when replayability becomes a native communication-layer capability, the application layer's handoff contracts need not themselves implement a fragile checkpoint tracker. The two are upper-lower layer cooperation, not repeated wheel-building.

4.4 Tension with the Sub-10-Millisecond Claim

Worth pointing out is an internal tension: replayable event streams mean persistence, audit, replayability, which naturally introduce extra write amplification and latency; while Adaptive-Bitmask-like claims pursue the extreme low latency of sub-10 milliseconds. These two goals are engineering-wise not necessarily compatible—it is hard to simultaneously demand "every event persisted and replayable" and "coordination latency pressed under 10 milliseconds." Real production systems often grade by task: long-horizon tasks needing strong audit and recoverability take replayable event streams; latency-sensitive, discardable lightweight coordination takes the bitmask-style fast path. UDOS's design corollary is: **the communication layer should not pursue a single latency target but grade channels by "whether replay is needed."**

4.5 Decision Latency vs Task Completion Time: Stop Confusing Them

Finally, a recurring confusion point must be nailed down. Numbers like sub-10 milliseconds refer to the latency of a **single coordination decision** (such as one bitmask vote, one negotiation reached); it absolutely does not equal the **end-to-end task** completion time. A multi-Agent task may need dozens of coordination rounds, each requiring LLM inference, and end-to-end time is minutes or even hours. Promoting "single coordination 10 milliseconds" into "million-Agent task completes in 10 milliseconds" is swapping a micro-benchmark for an end-to-end metric. The reason UDOS P1 insists on reporting "fan-in" and "critical-path rounds" separately is precisely to prevent this swap—single decision fast and whole critical path short are two things. The reason CNS simultaneously divides by T_{avg} also reminds: success rate and latency must be viewed together in one normalized metric, and looking at either alone misleads.

5 The Communication Layer's Real Bottleneck: Not Bandwidth

Combining the three threads, this paper's core judgment is: **the bottleneck of million-scale multi-Agent communication is not bandwidth but two things.**

Table 1 Repositioning the communication-layer bottlenecks

Intuitive bottleneck	Real bottleneck	Who solves it
Bandwidth insufficient	Whether it can deterministically replay after faults	RocketMQ-A2A replayable event streams
Too many messages	Whether coordination latency beats the decision cycle	Adaptive Bitmask bitmask (direction)
Protocols too messy	How to measure protocol scalability	AI Agent Protocols survey CNS

Bandwidth is the most easily misjudged bottleneck—people intuitively feel "million-Agent communication is bandwidth insufficient." But RocketMQ-A2A-like production solutions' real value is not "transmitting fast" but "resumable when broken, replayable when wrong, auditable throughout." This is the leap from "can transmit" to "can transmit trustworthily."

6 Dross, Failure Boundaries, and Falsifiable Conditions

6.1 Dross and circulated exaggeration

1. **Taking author claims as measurement.** Adaptive Bitmask's "85× compression / 8.2ms p99" is an author claim, not independently rechecked; directly writing "sub-10 milliseconds verified" is overreach.
2. **Taking coordination latency as end-to-end latency.** The bitmask presses negotiation decision latency, not task completion time; reading it as "million-Agent task completes in 10 milliseconds" is equivocation.
3. **Sovereign Negotiator / Ripple Effect specific numbers.** These two directions exist (layered negotiation, sensitivity-signal propagation), but their specific numbers like "87.6% token reduction / 41–100% coordination improvement" are not verified in this volume, and writing must mark [CITATION NEEDED], not as fact.

6.2 Failure boundaries

- This paper does not deny RocketMQ-A2A's production value; this paper only reminds "replayable event streams" solve **recovery and audit**, not **semantic-acceptance correctness** (still in P21/P27's domain).
- Sub-10-millisecond coordination is a **decision-latency** metric, and in semantic tasks where Agents need LLM inference, decision latency is often drowned by inference time, and bitmask gains may be insignificant.

6.3 Falsifiable conditions

- **C1**: if a RocketMQ-A2A-like replayable event stream in real production faults cannot achieve state consistency after checkpoint resume (replay results inconsistent with original execution), then the "replayable=auditable" engineering closed loop is weakened.
- **C2**: if Adaptive Bitmask's sub-10-millisecond claim in independent reproduction degrades rapidly with $\$N$ (such as at $\$N=5000$ $\$$ p99 far exceeds 10ms), then the "bitmask can press quadratic coordination" direction needs downgrading to a small- $\$N$ conclusion.

7 Hard Predictions (2026–2031)

- **F1**: over the next five years, production-grade multi-Agent systems' communication layer will take "replayable event streams + session isolation" as standard rather than competing on "peak bandwidth"; auditability replaces throughput as the main procurement metric.
- **F2**: sub-10-millisecond coordination-like claims, before independent reproduction, will be generally required by reviewers and procurement to give a p99 curve varying with $\$N$, rather than the best score at a single $\$N$ point.

8 Conclusion

P1 proved topology structure can press down the growth order of message organization; RocketMQ-A2A proved message middleware can build in fault recovery and audit; Adaptive Bitmask pointed to the direction of pressing coordination latency with compact representation; the AI Agent Protocols survey gave the ruler measuring all this. These four together are the complete engineering picture of "million-scale multi-Agent communication"—and its real bottleneck was never bandwidth, but whether it can deterministically replay after faults and whether coordination latency can beat the decision cycle.

For engineering, this means before deploying million Agents one must first ask three things: how are sessions isolated? How to resume when broken? Are replay results consistent? Only when the three questions have production-grade answers is "volume can be taken on" worth discussing. For research, it gives two falsifiable predictions (C1–C2): replayable event streams' state consistency and the bitmask's $\$N$ -scalability both need independent reproduction to confirm.

The communication layer's progress lies not in how fast it transmits, but in whether it can restart when broken and whether restarting is consistent.

References

2026 frontier materials (existence verified 2026-09-21 per EVIDENCE_LEDGER_v76.md)

1. Zhou, L., Zhang, S., et al. (2026). RocketMQ-A2A: Reliable Session-Level Replayable Event Streams for Large-Scale Multi-Agent Collaboration. ACM FSE 2026 Industry Papers Track.
2. (Adaptive Bitmask team) (2026). Adaptive Bitmask Protocols (npm package; sub-10ms direction true, compression ratio and p99 author claims, not independently rechecked).
3. (Survey Team) (2025). A Survey of AI Agent Protocols. arXiv:2504.16736.
4. Sovereign Negotiator Network (IEEE 2026; directional citation, specific numbers [CITATION NEEDED]).
5. Ripple Effect Protocol (directional citation, specific numbers [CITATION NEEDED]).

UDOS paper volume (this series, evidence grades marked in text)

1. UDOS v7.5.0: P1 "Million-Scale Extrapolation of Coordination Complexity of Layered Hybrid Topologies", P12 "Evidence-Grading-Driven Reproducible Agent Systems Engineering".
2. UDOS v7.6.0: P21 "End-to-End Evidence for Million-Scale Coordination and Its Extrapolation Boundary", P31 "Fail-Safe and Recovery".

Evidence Discipline and Reproduction Notes

- This paper is an engineering-survey/constraint-analysis paper with no new experiments; UDOS numbers return to P1/P12 and reports7/, evidence grades per each paper's labeling.
- RocketMQ-A2A's session-level replayable event streams are verified facts, "tens-of-millions-level sessions" product/paper-reported; Adaptive Bitmask's precise numbers are author claims, not independently rechecked; Sovereign Negotiator and Ripple Effect's specific numbers marked [CITATION NEEDED].
- This paper's predictions (F1–F2) and falsification conditions (C1–C2) can be post-hoc scored with public evidence during 2026–2031.